HAPAG LlYOD (LOGISTICS & DISTRIBUTION)

Navigating the Future: How Hapag-Lloyd Streamlined Fleet Management with a Centralized Dashboard

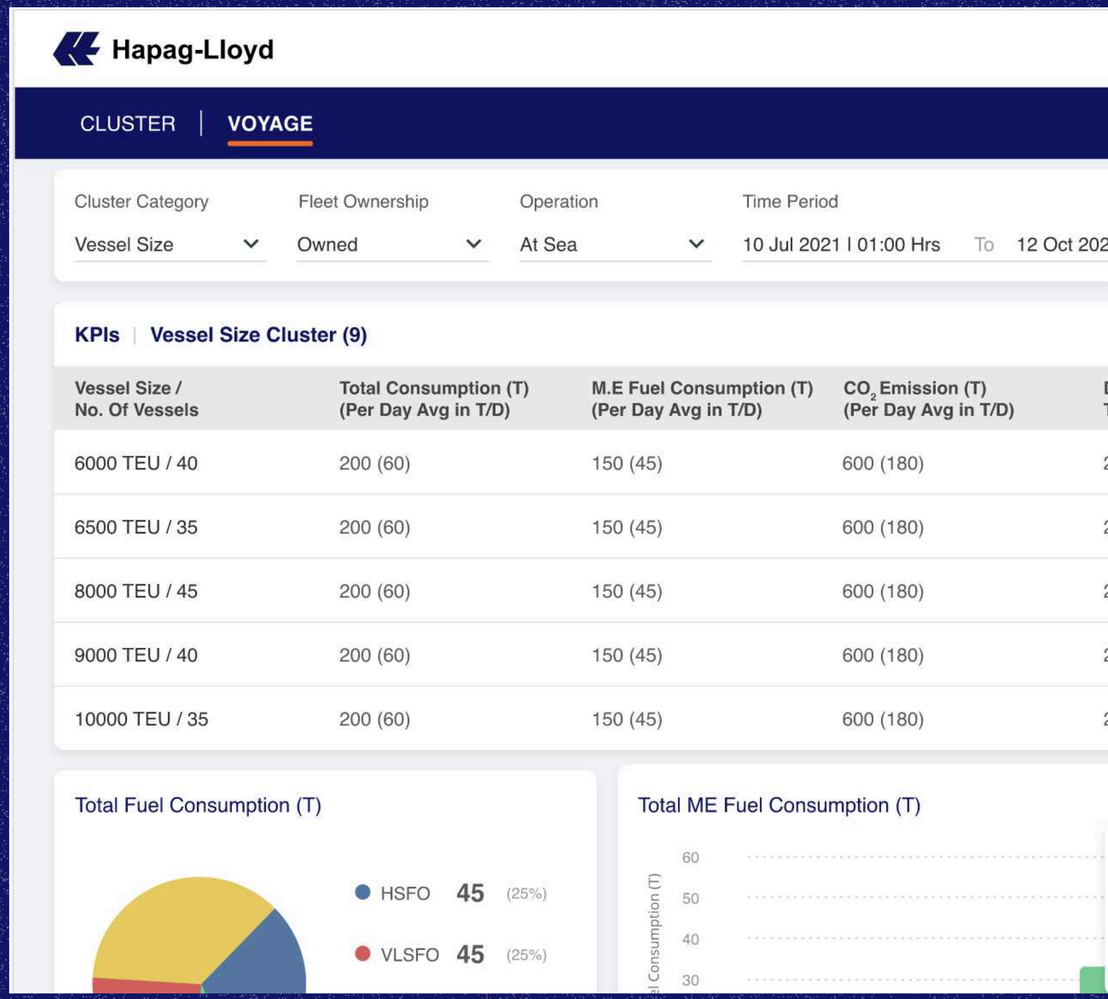


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Hapag-Lloyd

Hapag-Lloyd AG is a German international shipping and container transportation company. The roots of Hapag-Lloyd AG stretch far back into the 19th century, when the founding companies put their first ships to sea carrying general cargo and passengers to New York.



Hapag-Lloyd

253 vessels, 11.9 million TEU transport volume

Operates in more than 137 countries

My role

As the UX UI Designer, I designed clear, easy-to-use interfaces and visuals that help make better decisions, by focusing on what end users actually need.

Timeline

3 months

July 2022 - Sep 2022

Team Size

3 Designers + Manager

BACKGROUND

Problem Identification

Hapag-Lloyd needed a centralized dashboard to streamline fleet management by consolidating fragmented data. They sought real-time insights, KPI tracking, and support for decision-making to enhance operations and meet sustainability goals.

To create an integrated dashboard with

- Various KPIs (like fuel consumption, average weather)
- Data about the ships & their voyages
- Comparisons between different KPIs



BACKGROUND

Terminology

1 Voyage

A Mumbai → B Dubai → C Mumbai

1 Leg of Voyage

A Mumbai → B Dubai

*a voyage can have 3-5 legs

At Sea



The operation when a ship is at sea.

At Port



The operation when a ship is at port.

Maneuvering



When a ship is arriving, departing a port or crossing canals & traffic zones.

Research Sprint

After **rigorous discussions** with the stakeholders we understood more about the fleet and logistics data in detail which included **key indicators** such as CO2 emissions, Engine Fuel Consumption, Average Speed, Operations, Time period, Fleet ownership.

RESEARCH

Persona

Based on **research**, we created a **persona** to represent our **ideal target audience** which gave us an idea of their **motivations and frustrations**

Smith Jones

35, Chief Officer

Bio

Smith Jones is a Chief Officer whose job concerns ship’s safety, ship’s security & environmental protection issues.

Goals

- To be aware of all the periodical data regarding ships, like Co2 emissions etc.
- To get updates about important tasks.

Frustrations

- Unable to keep a track of data for ships, vessels & voyages.
- Missing important alerts or unaware of urgent tasks that might need tending.



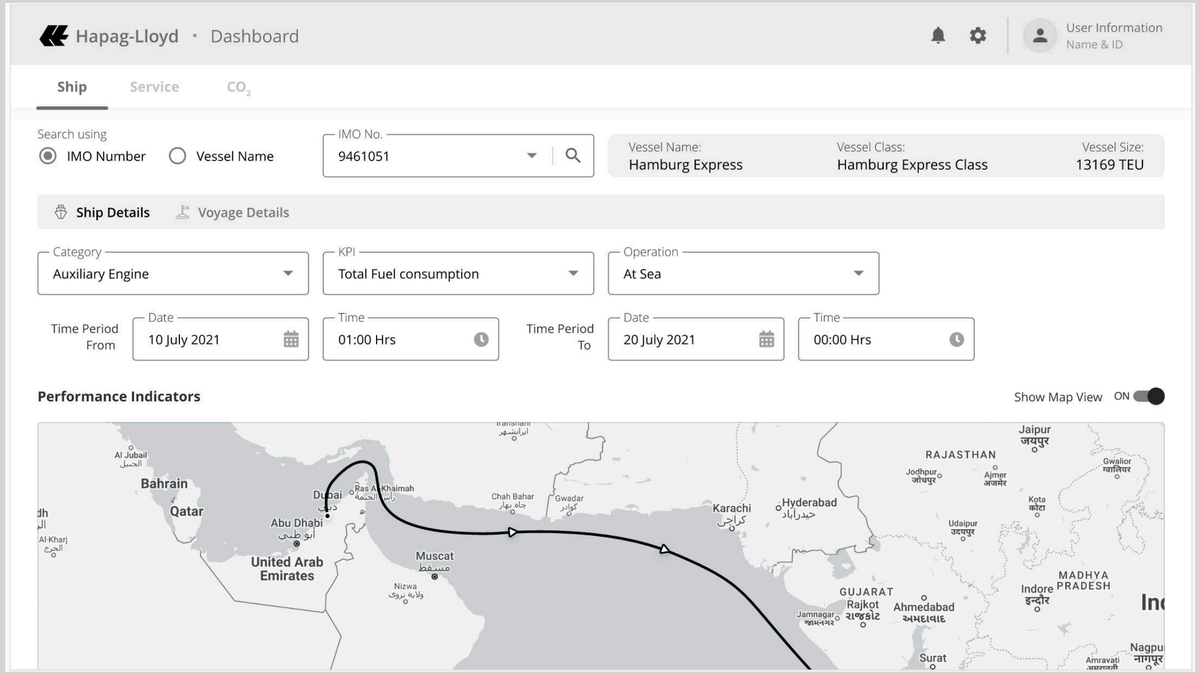
INITIAL CONCEPT

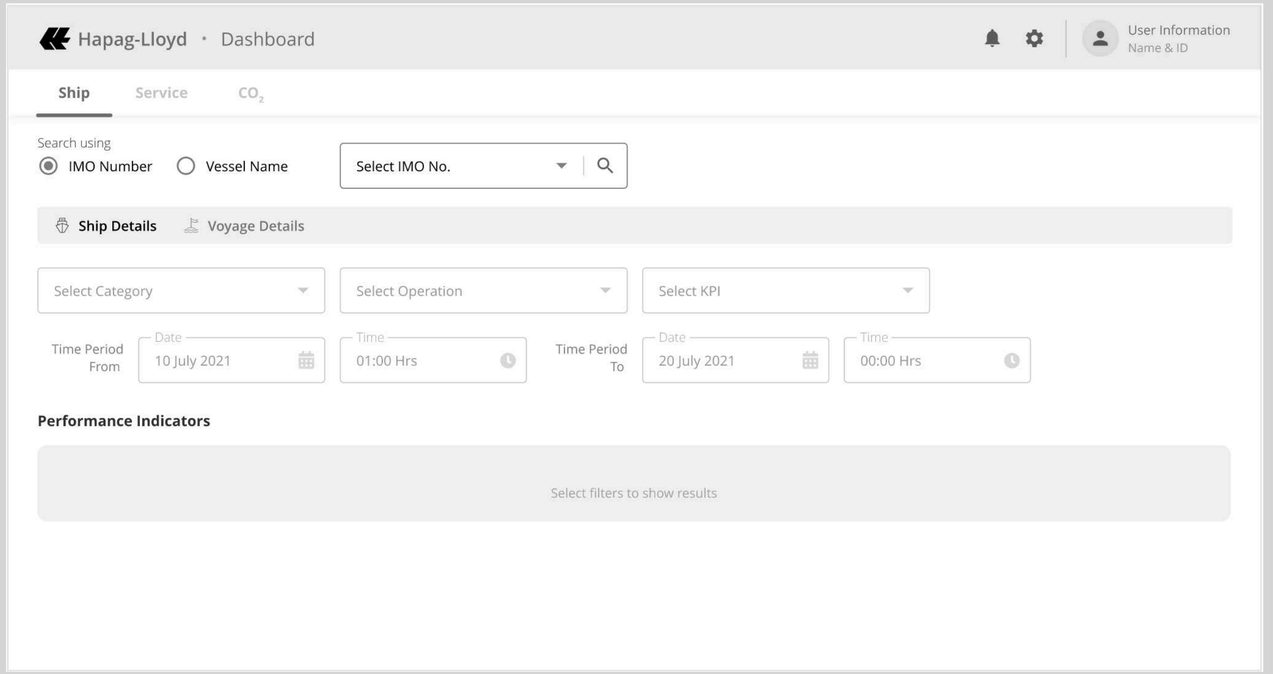
High fidelity wireframes

Several brainstorming sessions led us to creating initial low-fidelity wireframes to get started with a direction for our idea of the solution

Key design considerations included:

- Simplified navigation with clear data hierarchies
- Color-coded KPIs for intuitive understanding

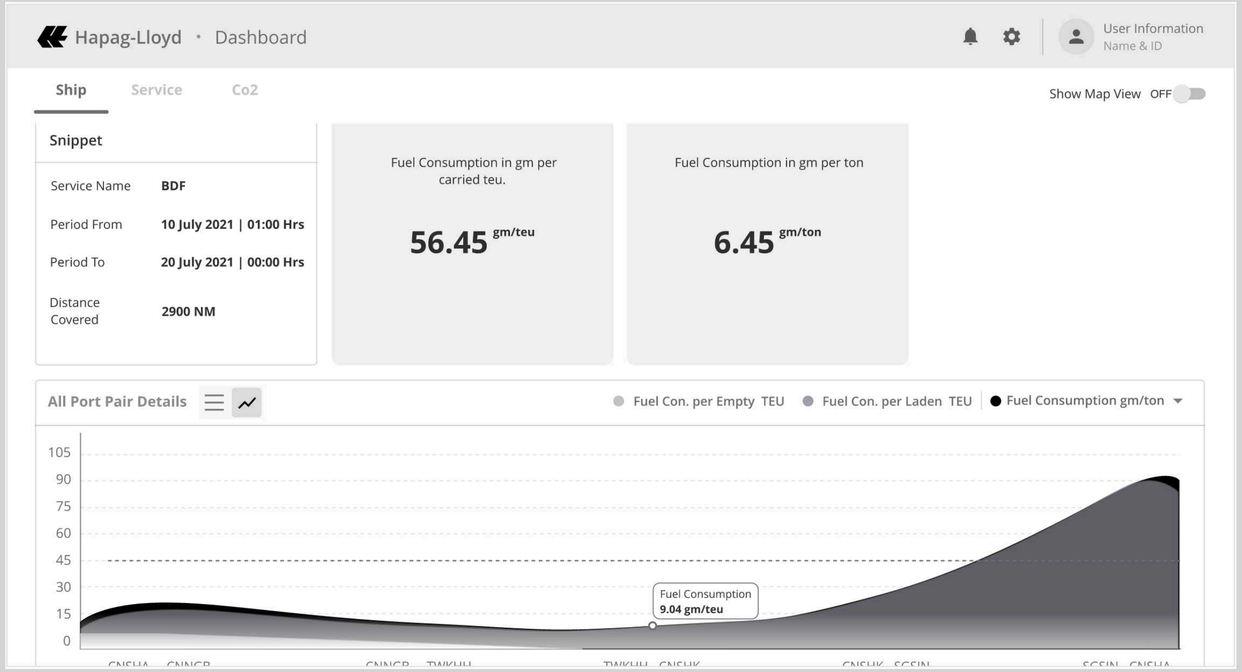


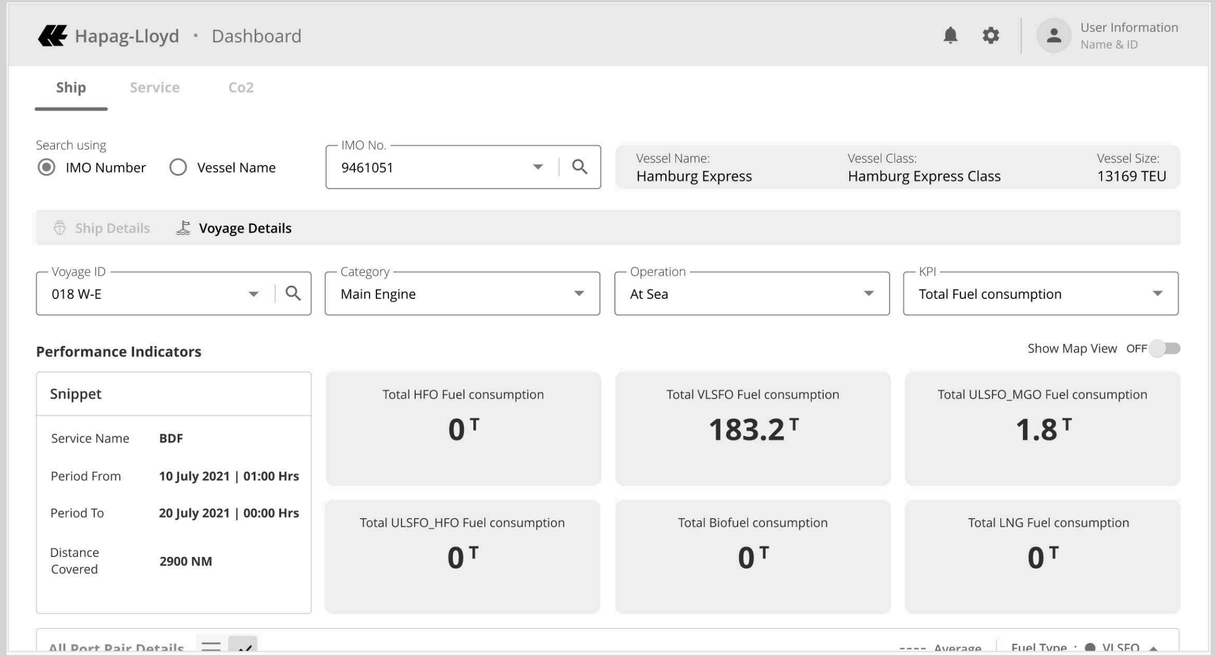


Other high fidelity wireframes

Key design considerations included:

- Customizable filters for relevant data extraction
- Visual and tabular views to reduce cognitive load





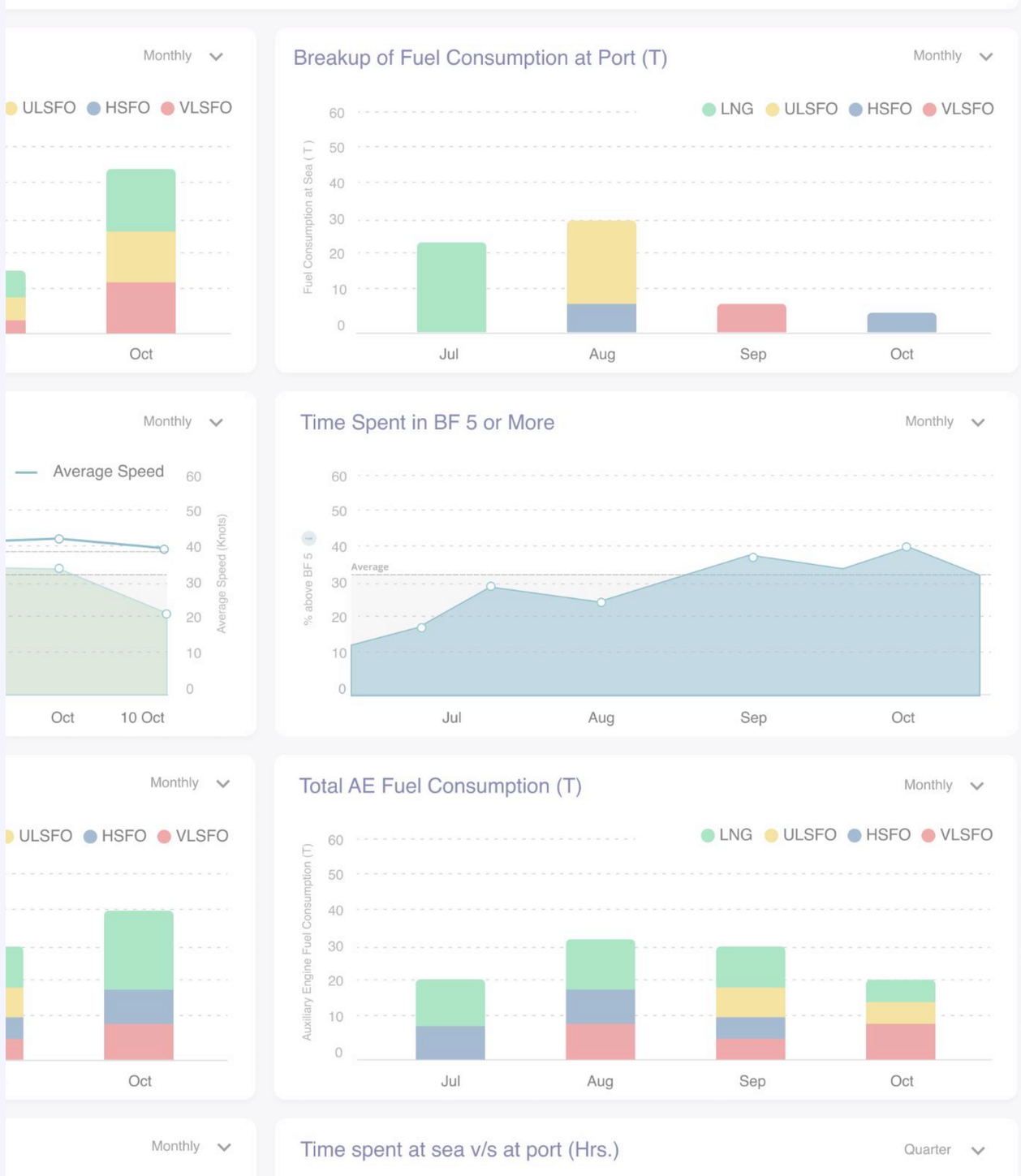
After multiple iterations

And final approval

The Solution

To create an integrated dashboard with

- User-friendly, easy to navigate and requires minimum steps to do tasks
- Information hierarchy used to highlight important data
- Data visualisation: Different type of graphs to avoid cognitive load & monotony



The Design

Filter options of category and operations to see the results accordingly

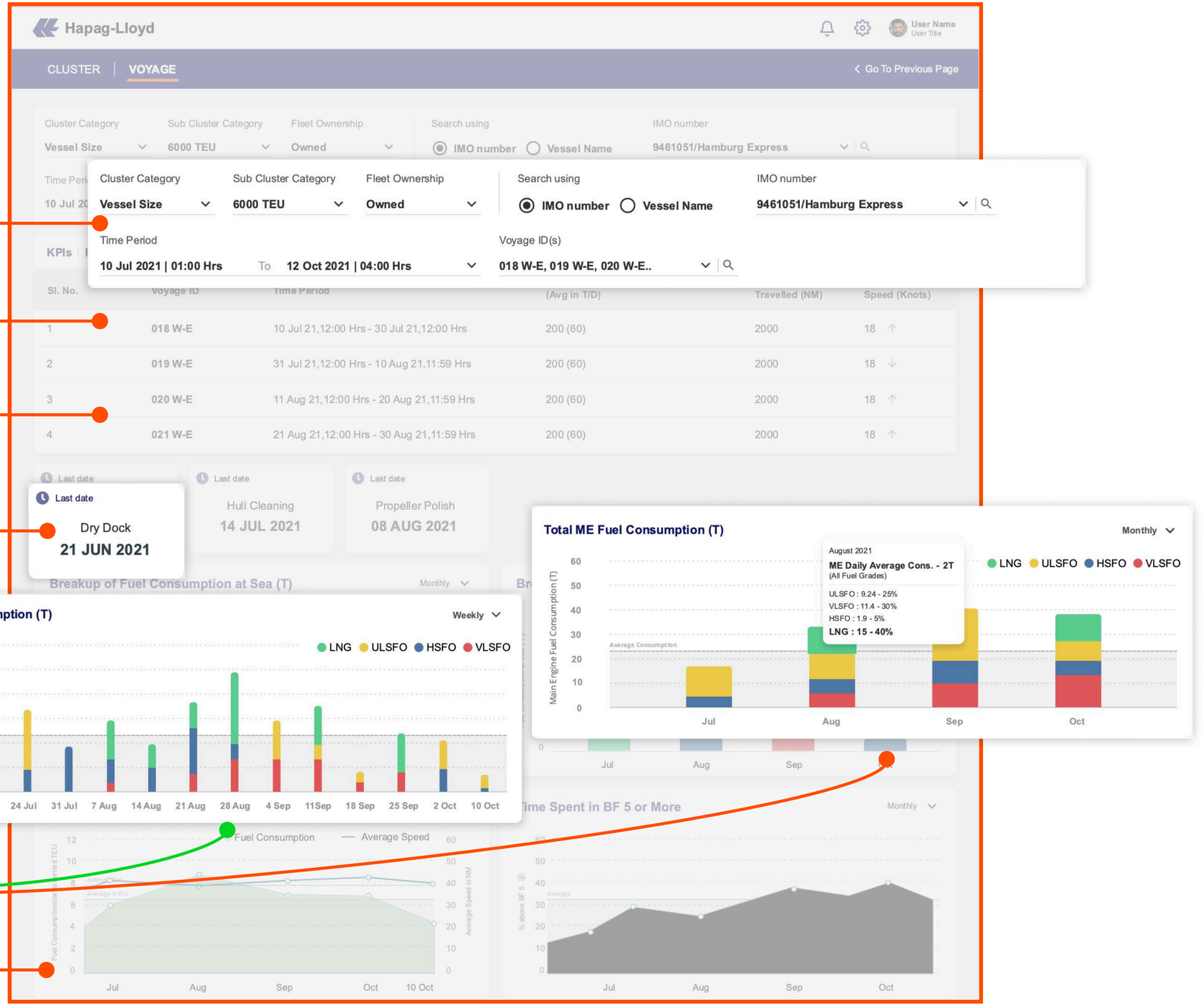
Table view (basic information)

A design system built by keeping the essence of the brand guidelines

Highlighted information that are required to see briefly

Color coded KPIs and curated data according to their preference of weekly/ monthly/ quarterly

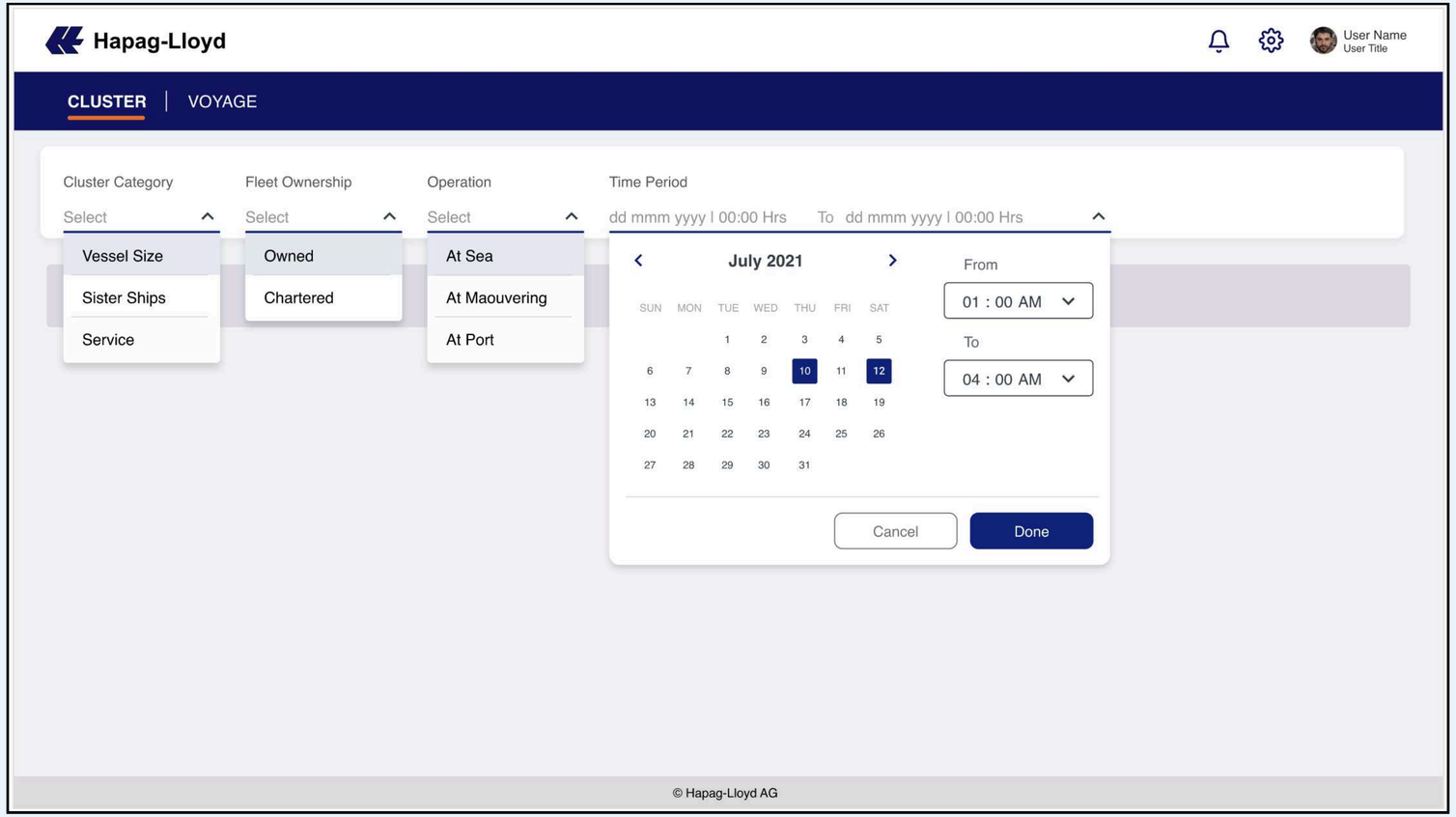
Graphical view (KPIs)



The Design (continued)

Dashboard Design

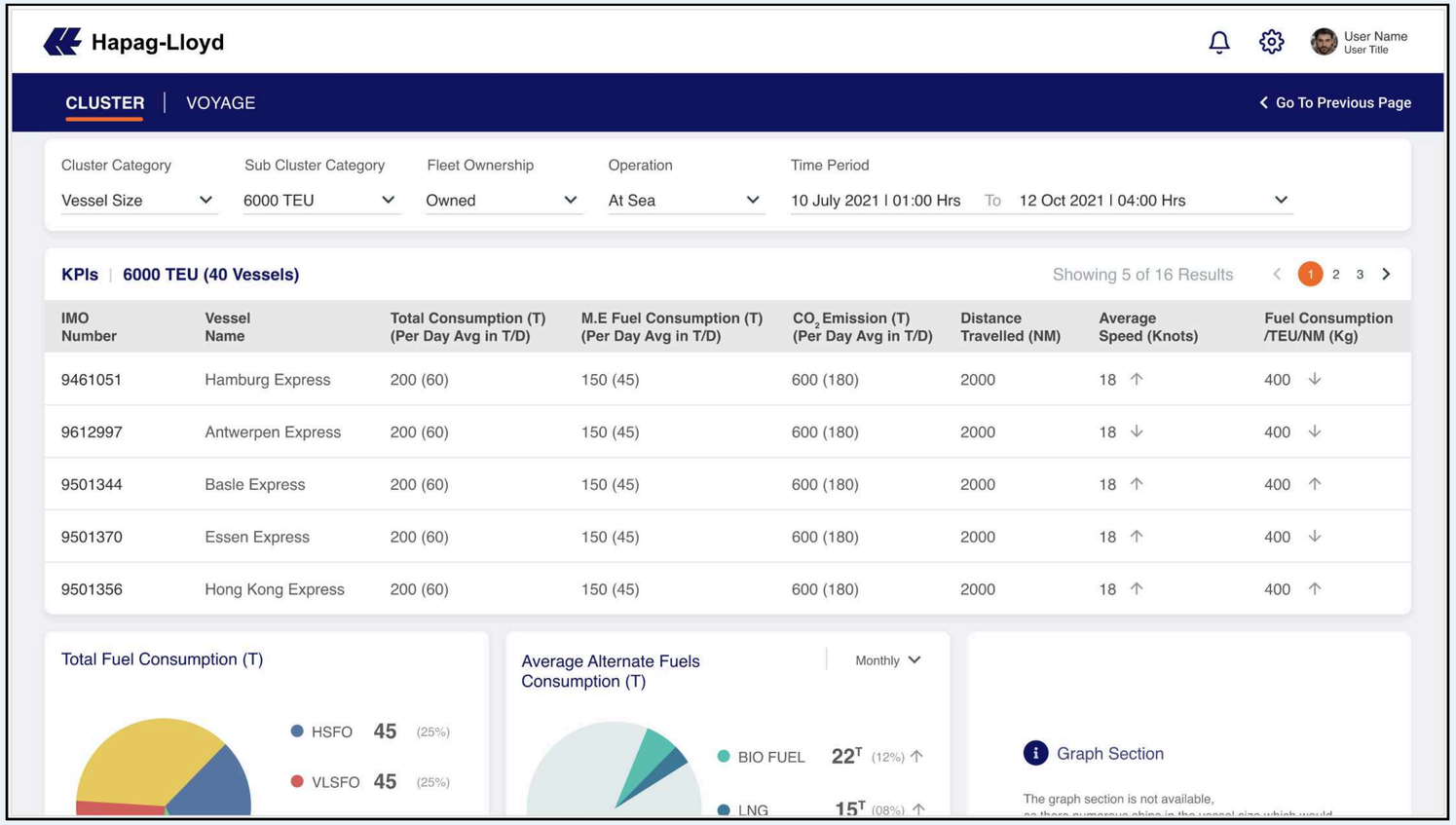
- The Landing page has two parts **“Cluster”** and **“Voyage”**.
- The Cluster tab allows users to choose various filtering parameters like **cluster category, fleet ownership, operation and time period.**



The Design (continued)

Cluster Dashboard

- Upon choosing the cluster category as vessel size, fleet data is displayed in the form of tables and graphs.
- The table includes vessel details and respective KPIs along with inline pagination to access the extensive data.



Overall Impact



50% REDUCTION in Decision-Making Time

Faster access to insights allowed quicker responses to operational challenges.



40% IMPROVEMENT In Operational Efficiency

Reduced time spent on data consolidation and analysis.



20% REDUCTION in CO2 Emissions

Enhanced tracking and monitoring facilitated sustainable operational practices.



80% INCREASE in User Satisfaction

Improved user experience through intuitive design and real-time data visualization.

Thank you